

Final Examination

Duration: 120 minutes

- Each question is worth **2 points**, for a total of **100 points**. Please circle your answer directly on this paper. **Example:**

How much did you like the "Introduction to Computer Science" course?

- a. Not at all b. Somewhat c. Very much ☒ d. Extremely
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- Q1.** A "Greedy Algorithm" is considered a heuristic approach because:
- a. It looks ahead at all possible future moves.
 - b. It always picks the best immediate (local) option without considering the long-term result.**
 - c. It is only used in financial applications.
 - d. It never fails to find the global optimum.
- Q2.** If a neural network performs perfectly on training data but very poorly on new, unseen test data, it is likely experiencing:
- a. Overfitting**
 - b. Underfitting
 - c. Vanishing Gradients
 - d. High Bias
- Q3.** In the RGB color model, if you set Red, Green, and Blue to their maximum values (e.g., 255, 255, 255 in 8-bit), what color is produced?
- a. Black
 - b. Grey
 - c. White**
 - d. Transparent
- Q4.** Which of the following is the defining characteristic of Unsupervised Learning?
- a. The model requires a "teacher" to provide the correct answers during training.
 - b. It is primarily used for predicting continuous numerical values like house prices.
 - c. It uses a reward-based system to learn through trial and error.
 - d. The algorithm works with data that has no predefined labels or output categories.**
- Q5.** What is "Anti-aliasing" used for in computer graphics?
- a. To increase the speed of the rendering engine.
 - b. To smooth out the "jagged" edges (jaggies) of lines and shapes.**

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- c. To add shadows to a 3D scene.
 - d. To compress the file size of an image.
- Q6.** "Ray Tracing" is a rendering technique that is different from Rasterization because it:
- a. Ignores light sources to save processing power.
 - b. Only works with 2D images.
 - c. Always runs faster than Rasterization.
 - d. **Traces the path of individual light rays to simulate realistic reflections, shadows, and refractions.**
- Q7.** In the Phong Shading model, which component represents the "shiny" bright spot on an object caused by a light source?
- a. Ambient
 - b. Diffuse
 - c. **Specular**
 - d. Emissive
- Q8.** Which of the following is a key advantage of using a DBMS over a traditional file-processing system?
- a. Increased data redundancy
 - b. Dependency between programs and data
 - c. **Reduced data inconsistency**
 - d. Lower hardware requirements
- Q9.** Which SQL command is part of the Data Definition Language (DDL) or change the database structure.
- a. **CREATE**
 - b. SELECT
 - c. INSERT
 - d. UPDATE
- Q10.** Which of the following best describes a Database Management System (DBMS)?
- a. A hardware component used to store data files.
 - b. **Software that enables users to define, create, maintain, and control access to the database.**
 - c. A programming language used only for mathematical calculations.
 - d. A set of raw facts without any context.
- Q11.** A "Many-to-Many" relationship between two entities (e.g., Students and Courses) is typically implemented in a relational database by:
- a. Adding a foreign key to the "Student" table.

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- b. Adding a foreign key to the "Course" table.
- c. Creating a third table, often called a junction or associative table.
- d. Merging both entities into a single table.

Q12. In a Relational Model, a "tuple" refers to a:

- a. Column
- b. Table
- c. Key
- d. Row

Q13. Which of the following is a NoSQL database type?

- a. MySQL
- b. Oracle
- c. PostgreSQL
- d. MongoDB

Q14. Which of the following is considered an "Abstract Data Type" (ADT)?

- a. int
- b. char
- c. Stack
- d. double

Q15. What is the main difference between Raster and Vector graphics?

- a. Raster graphics use mathematical paths; Vector graphics use a grid of pixels.
- b. Raster graphics are made of pixels; Vector graphics are defined by mathematical formulas and paths.
- c. Vector graphics lose quality when resized; Raster graphics do not.
- d. Raster graphics are only used for 3D; Vectors are only used for 2D.

Q16. In Object-Oriented Programming, which mechanism is most used to achieve data abstraction?

- a. Inheritance
- b. Pointers
- c. Classes and Objects
- d. Loops

Q17. Which data abstraction follows the Last-In, First-Out (LIFO) principle?

- a. Array
- b. Stack
- c. Linked List
- d. Memory Buffer

Q18. In the context of a "Car" object, which of the following is an abstraction?

- a. The steering wheel and pedals.

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- b. The internal combustion engine's piston timing.
- c. The wiring diagram of the dashboard.
- d. The chemical composition of the fuel.

Q19. Which of the following is the primary advantage of a Linked List over an Array?

- a. Faster access to random elements
- b. Lower memory overhead per element
- c. **Dynamic size and efficient insertions/deletions**
- d. Better cache locality

Q20. Which of the following is used to uniquely identify a row within a table?

- a. Foreign Key
- b. **Primary Key**
- c. Secondary Key
- d. Composite Attribute

Q21. Which operation is used to add an element to the top of a Stack?

- a. **Push**
- b. Pop
- c. Enqueue
- d. Append

Q22. A "Foreign Key" is used to:

- a. Uniquely identify a row within its own table.
- b. **Establish and enforce a link between the data in two tables.**
- c. Encrypt sensitive data in a table.
- d. Automatically sort the table in alphabetical order.

Q23. A Queue data structure is best described as:

- a. LIFO (Last-In, First-Out)
- b. FIFO (First-In, First-Out)
- c. LILO (Last-In, Last-Out) D)
- d. **Both B and C**

Q24. What is the maximum number of children a node can have in a Binary Tree?

- a. 1
- b. **2**
- c. 3
- d. Unlimited

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Q25. In a Queue, where does the "Dequeue" operation typically occur?

- a. At the Rear
- b. At the Front
- c. In the Middle
- d. At any random position

Q26. Which traversal of a Binary Search Tree (BST) yields the elements in sorted ascending order?

- a. Pre-order
- b. Post-order
- c. In-order
- d. Level-order

Q27. What do "Weights" represent in an ANN?

- a. The physical size of the neural network.
- b. The speed at which the network processes data.
- c. The strength or importance of the connection between two neurons.
- d. The number of training examples in a dataset.

Q28. What is a "Circular Linked List"?

- a. A list where every node points to the head.
- b. A list where the last node points back to the first node.
- c. A list that can only store numeric data.
- d. A list that is stored in a circular buffer.

Q29. Which data structure is most efficient for implementing "Undo" functionality in a text editor?

- a. Stack
- b. Queue
- c. Binary Tree
- d. Linked List

Q30. What is the time complexity of searching for an element in a balanced Binary Search Tree (BST)?

- a. $O(1)$
- b. $O(n)$
- c. $O(\log n)$
- d. $O(n^2)$

Q31. In a Binary Tree, a node that has no children is called a:

- a. Root node
- b. Leaf node

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- c. Interior node
- d. Parent node

Q32. What happens during a "Stack Overflow"?

- a. The stack is empty and a pop is attempted.
- b. Two elements are pushed at the same time.
- c. The data in the stack is corrupted.
- d. The stack exceeds its allocated memory/capacity.

Q33. In a Binary Search Tree, where would a value smaller than the root be placed?

- a. In the left subtree
- b. In the right subtree
- c. As the new root
- d. It depends on the height of the tree

Q34. Which operation is generally $O(1)$ for a Stack implemented with a Linked List?

- a. Push
- b. Pop
- c. Peek
- d. All of the above

Q35. What is the "Height" of a tree?

- a. The total number of nodes in the tree.
- b. The number of leaf nodes.
- c. The number of children the root has.
- d. The number of edges on the longest path from the root to a leaf.

Q36. What is the process of converting source code into machine code all at once before execution?

- a. Compilation
- b. Interpretation
- c. Transpilation
- d. Debugging

Q37. What does "syntax" refer to in a programming language?

- a. The meaning of the code.
- b. The set of rules that defines the combinations of symbols that are considered correctly structured programs.
- c. The speed at which the code runs.
- d. The library of pre-built functions.

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Q38. Which of the following is a "low-level" programming language?

- a. Python
- b. C++
- c. **Assembly**
- d. Ruby

Q39. Which of the following describes "Recursion"?

- a. A loop that never ends.
- b. **A function that calls itself.**
- c. A method to encrypt data.
- d. A way to store multiple values in one variable.

Q40. Which language is primarily used for Data Science and Machine Learning due to its extensive libraries?

- a. C#
- b. PHP
- c. **Python**
- d. Swift

Q41. Which of the following is an example of a "Greedy" algorithm?

- a. Merge Sort
- b. **Dijkstra's Shortest Path**
- c. Binary Search
- d. Fibonacci calculation via recursion

Q42. A "P" class problem in complexity theory refers to problems that can be solved in:

- a. Probabilistic time
- b. **Polynomial time**
- c. Parallel time
- d. Partial time

Q43. Which of the following is a characteristic of Dynamic Programming?

- a. It always uses the most memory.
- b. It makes the locally optimal choice at each stage
- c. It only works on sorted arrays.
- d. **It solves each sub-problem only once and stores the result (memorization).**

Q44. Which protocol is used for securely transferring files over a network?

- a. HTTP

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- b. FTP
- c. SFTP
- d. Telnet

Q45. What is the standard port number used for HTTPS traffic?

- a. 21
- b. 25
- c. 80
- d. 443

Q46. Which of the following protocols is used to send emails?

- a. SMTP
- b. POP3
- c. IMAP
- d. HTTP

Q47. Which hardware address is unique to every Network Interface Card (NIC)?

- a. IP Address
- b. MAC Address
- c. Port Address
- d. Gateway Address

Q48. What does the term "Open Source" mean?

- a. The software is free to buy.
- b. The software can only be used on Linux.
- c. The software is always beta and unfinished.
- d. The source code is available for anyone to inspect, modify, and enhance.

Q49. What does the "Operating System" act as an intermediary between?

- a. The CPU and the GPU
- b. The user and the computer hardware
- c. The internet and the local network
- d. The monitor and the keyboard

Q50. In the context of cybersecurity, what does "Encryption" do?

- a. Converts information into a secret code to prevent unauthorized access.
- b. Speeds up the download process.
- c. Deletes old files.
- d. Repairs a broken hard drive.